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Session Objectives

1. Correlate ECG parameters with the physiology of the myocardium and cardiac conduction system.
2. Identify and understand lead placement of limb & precordial leads (Standard 12-lead ECG).
3. Define vectors in relation to the electrical potential amplitude and direction.
4. Apply Einthoven’s triangle to recognize and assess ECG characteristics.

First Aid (2025): 298

Source: Course Syllabus

What is an ECG?

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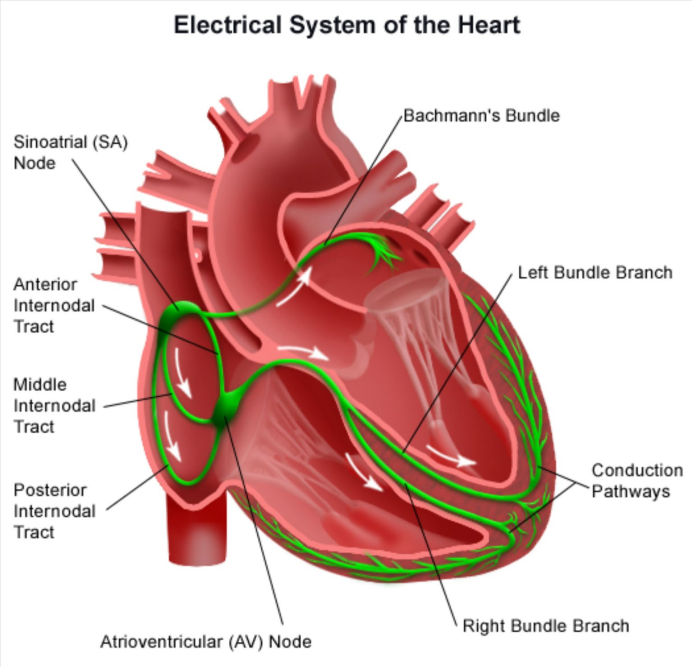
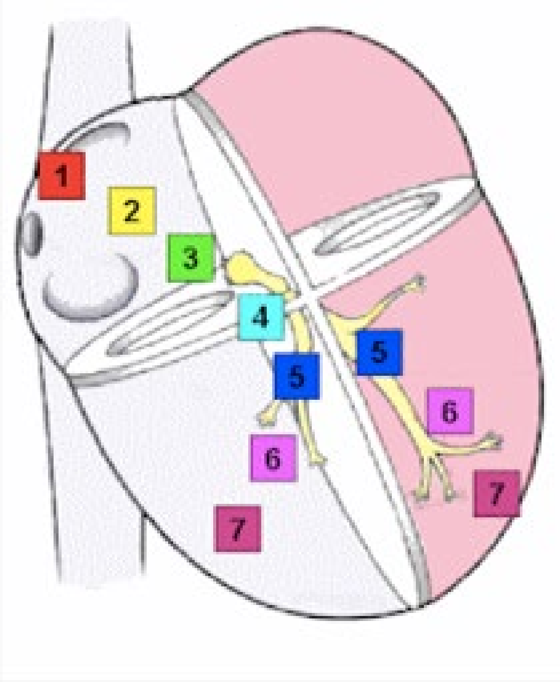
- An electrocardiogram (EKG or ECG) is a graphic representation of the electrical activity of the heart
- Comprised of 12 leads placed on the precordium (or chest) and all 4 limbs
- Can detect changes in electrical potential produced in successive areas of the myocardium during the cardiac cycle

Conduction System

- Conduction pathway: SA node → atria → AV node → bundle of His → right and left bundle branches → Purkinje fibers → ventricles
- Pacemaker rates: SA node (60 - 100 bpm), AV node (40 - 60 bpm), bundle of His/Purkinje/ventricles (25 - 40 bpm)
- Speed of conduction: **His-Purkinje** > **Atria** > **Ventricle** > **AV** (**H**e **P**arked **At** **Ven**tura **AV**e)
- SA node and AV node receive their blood supply from the right coronary artery (RCA)



Pacemaker Speeds & Conduction Velocity



Normal Activation Sequence	Structure	Conduction velocity (m/sec)	Pacemaker rate (beats/min)
1	SA node	< 0.01	60 – 100
2	Atrial myocardium	1.0 – 1.2	None
3	AV node	0.02 – 0.05	40 – 55
4	Bundle of His	1.2 – 2.0	25 – 40
5	Bundle branches	2.0 – 4.0	25 – 40
6	Purkinje network	2.0 – 4.0	25 – 40
7	Ventricular myocardium	0.3 – 1.0	None

FAST!!

Purkinje System

⇓

Atrial Muscle

⇓

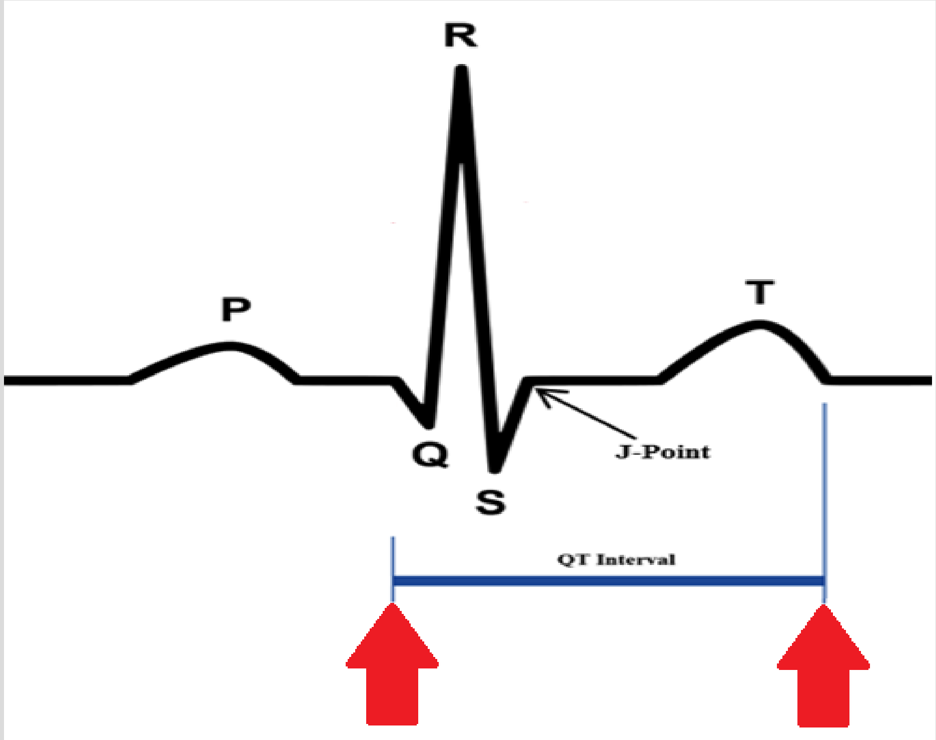
Ventricular Muscle

⇓

AV Node

...slow...

QT Interval



- QT interval- ventricular depolarization, mechanical contraction of the ventricles, ventricular repolarization (aka ventricular systole)
- Generally less than half of the R-R interval
- Prolonged QT increases risk of ventricular arrhythmias
  - Congenital Long QT Syndrome
  - Electrolyte imbalances
  - **Drug induced**

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Objective 1 Question:

Which of the following represents ventricular repolarization?

a. QRS complex

b. P wave

c. T wave

Objective 1 Question:

A 29 year old male with depression presents to your office with a family history of sudden death of his uncle and grandfather. He has tried multiple SSRI's and SNRI's to no success. He is interested in starting imipramine. On EKG, you will be most worried about the

a. PR interval

b. QT interval

Different presentations of an ECG

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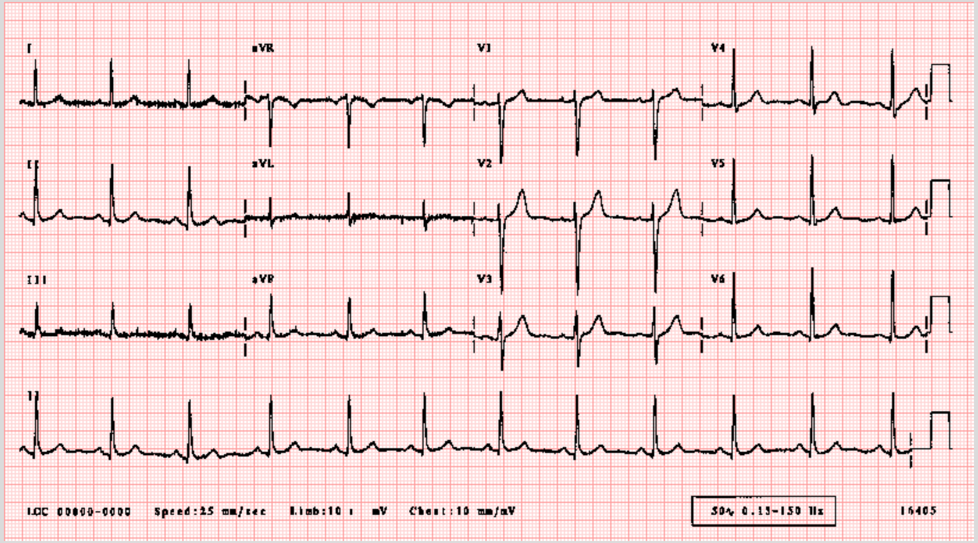
- If you count 12 leads = 12-lead ECG



- If you only see 1 lead = Rhythm Strip

Objective 2 Question:

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Which of the following can NOT be identified by an ECG?

a. STEMI (myocardial infarction)

b. Primary Biliary Cholangitis

c. Arrhythmias such as Atrial fibrillation

Objective 2 Question:

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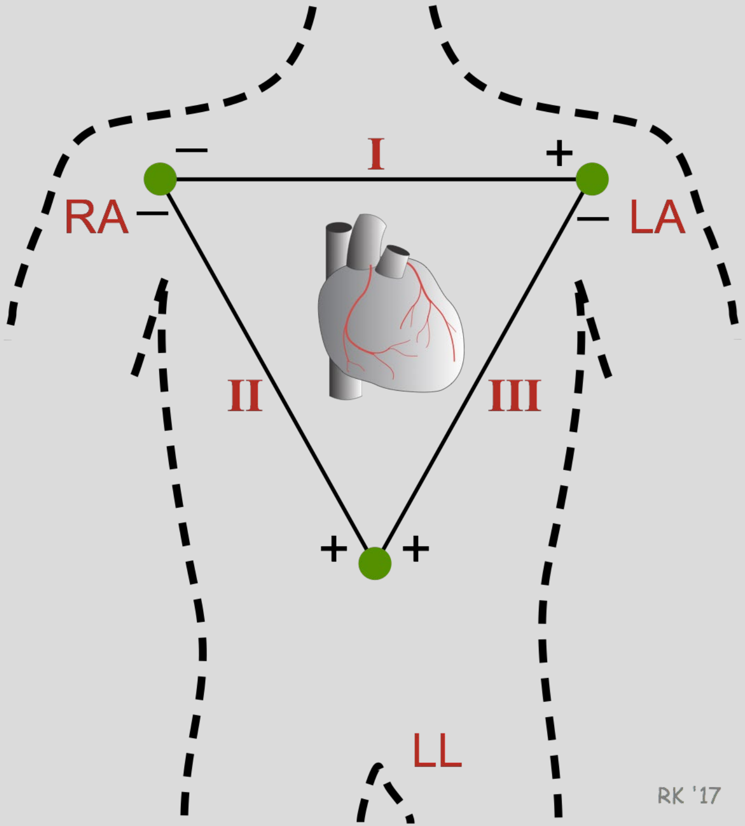
An 84 year old female with a PMHx of uncontrolled HTN and Diabetes presents with crushing chest pain radiating down the arm for the past 40 minutes. EMS has provided her with nitroglycerin, aspirin, oxygen, and morphine on route that has mildly assisted the patient. The EKG is not attached on arrival, and your attending asks you to place a white lead on the patient. You place it on the Right arm. Will your attending be disappointed in you?

Optional Answers: **1. No, my attending is proud!**

2. Back to studying limb leads…

Einthoven's Triangle: I, II, III



- Lead I- potential between RA and LA
- Lead II- potential between RA and LL
- Lead III- potential between LA and LL

Mnemonic

- I=1 L=RA to LA (Right Arm to Left Arm)
- II=2 L=RA to LL (Right Arm to Left Leg)
- III=3L=LA to LL (Left Arm to Left Leg)

Objectives 3 & 4 Questions:

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Which of the following are precordial leads?

a. I

b. avF

c. V4

d. none of the above

Objectives 3 & 4 Questions:

A 74 year old male presents to the ER complaining of severe substernal chest pain. You quickly triage and see that this patient may be presenting with a STEMI so you order a stat EKG. Your attending asks you to identify the axis. Which pair of leads is most suitable for this purpose?

a. avF & II

b. avR & avF

c. I & avF

Summary Slide *w/ some extra information*

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- P wave- atrial depolarization. Atrial repolarization is masked by QRS complex
- PR interval- time from start of atrial depolarization to start of ventricular depolarization (normally < 200 msec). AV node conduction in PR interval prolongations
- QRS complex- ventricular depolarization (normally < 120 msec).
- T wave- ventricular repolarization. Hypo/Hyperkalemia. Recent Myocardial Infarction.
- QT interval- ventricular depolarization, mechanical contraction of the ventricles, ventricular repolarization. Normally half the R-R interval.
- ST segment- isoelectric, ventricles depolarized. Elevations (transmural), depression (subendocardial)
- TP segment- true isoelectric point, useful to assess ST segment changes

How I would approach the lecture:

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2. Identify and understand placement of limb & precordial leads (Standard 12 lead).

a. What are the leads you attach for a 12 lead?
b. What does a 12 lead EKG look like?
c. R wave progression as an introduction to ventricular depolarization as it goes across the precordium

How I would approach the lecture:

3. Define vectors in relation to the amplitude and direction.
   a. Pathway of electricity through the heart & orientation of the heart
   b. Pacemaker rates of the heart
   c. What is a vector?
   d. What is/ How is a vector calculated for the heart?